

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- **Key Features:**
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data

Analysis using Python We began with data preparation and cleaning in Python:

- Data Loading: Imported the dataset using pandas.
- Initial Exploration: Used `df.info()` to check structure and `.describe()` for summary statistics

Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
Yes	Yes	14	Venmo	Fortnightly
Yes	Yes	2	Cash	Fortnightly
Yes	Yes	23	Credit Card	Weekly
Yes	Yes	49	PayPal	Weekly
Yes	Yes	31	PayPal	Annually

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Feature Engineering:** ○ Created age_group column by binning customer ages. ○ Created purchase_frequency_days column from purchase data.
- **Data Consistency Check:** Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL

(Business Transactions) We performed structured analysis in PostgreSQL to answer key business questions:

1- What is the total revenue generated by Male and Female customers?

	gender	revenue
1	Male	157890
2	Female	75191

2 - Which customer used a discount but still spent more than the average purchase amount?

	customer_id	purchase_amount
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
14	33	67
15	35	91
16	37	69
17	40	60

3- Which are the top 5 products with the highest average review rating?

	item_purchased	Average product Rating
1	Gloves	3.86277372262774
2	Sandals	3.84465408805031
3	Boots	3.81888111888112
4	Hat	3.80130718954248
5	Skirt	3.78535031847134

4- Compare the average purchase amount between standard and express shopping?

	subscription_status	total_customers	avg_spend	total_revenue
1	No	2847	59	170436
2	Yes	1053	59	62645

5- Do subscribed customers spend more? Compare average spend and the total revenue between subscribers and non-subscribers

	subscription_status	total_customers	avg_spend	total_revenue
1	No	2847	59	170436
2	Yes	1053	59	62645

6- Which 5 products have the highest percentage of purchase with discounts applied?

	item_purchased	discount_rate
1	Hat	50.000000
2	Sneakers	49.660000
3	Coat	49.070000
4	Sweater	48.170000
5	Pants	47.370000

7- What is the revenue contribution of each age group?

	age_group	total_revenue
1	Young Adult	62143
2	Middle-Aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.

